Hallucination

Limitations of LLMs about Knowledge

Knowledge cutoff

- LLMs don’t know about any events that happened after their training
- LLMs don’t have any knowledge about private or confidential information that they have not encountered during training

Hallucinations

- LLMs are trained to generate realistic-sounding or convincing text
- But the generated text may be nonetheless wrong
  - instead of admitting that it lacks the base facts in its training.

How to Resolve Knowledge Cutoff?

Augmenting LLMs w/ external resources

- Retrieval Augmented Generation (RAG): supplement the LLM’s internal knowledge by external information
  - retrieving facts from an external documents or a knowledge base to ground LLMs on the most accurate, up-to-date information and to give as context into LLMs' generative process.

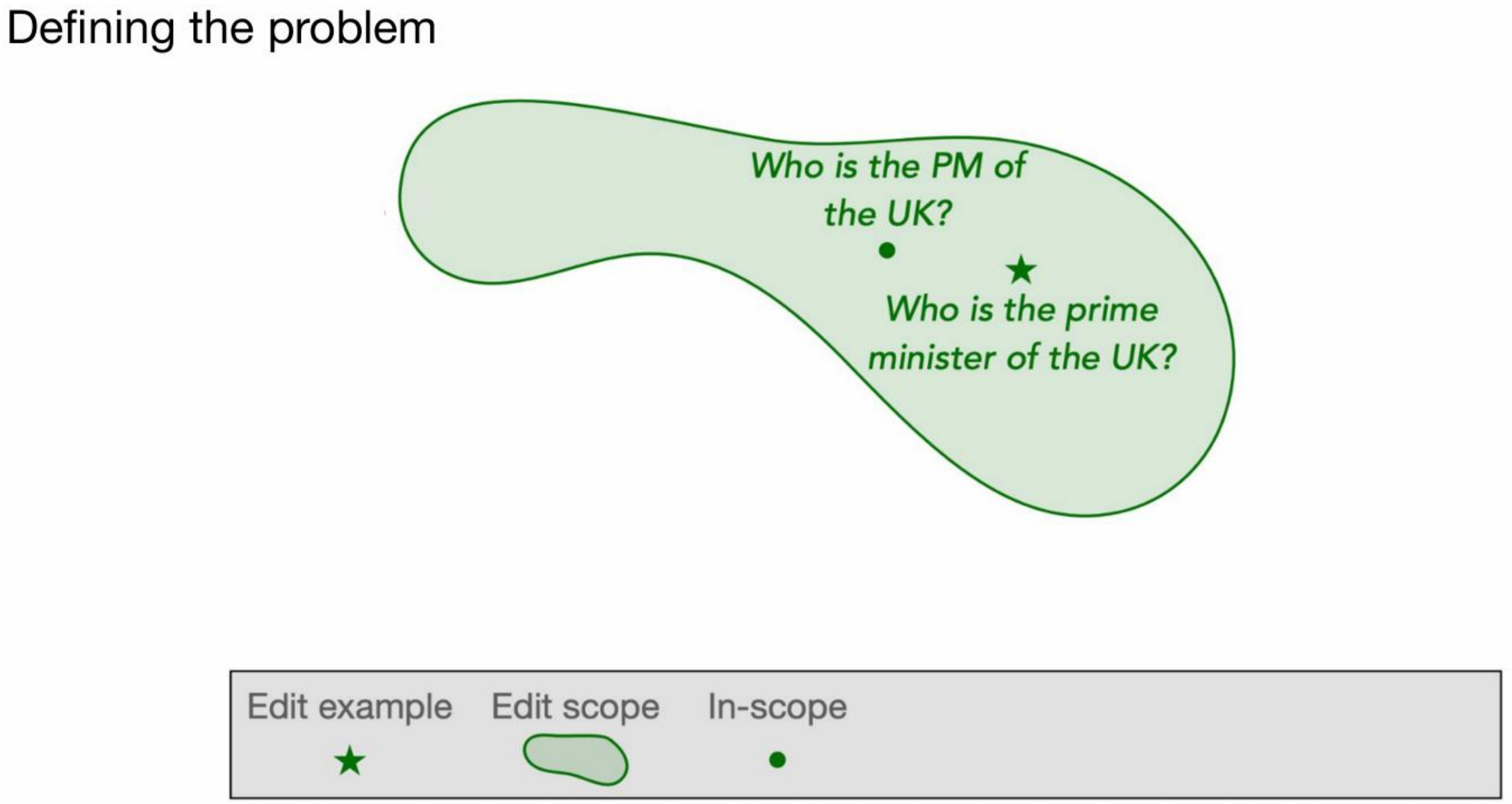
Knowledge Editing

- Factuality purpose: Can knowledge be edited?
- Privacy purpose: Can sensitive or private information be deleted from LLMs (unlearning)?

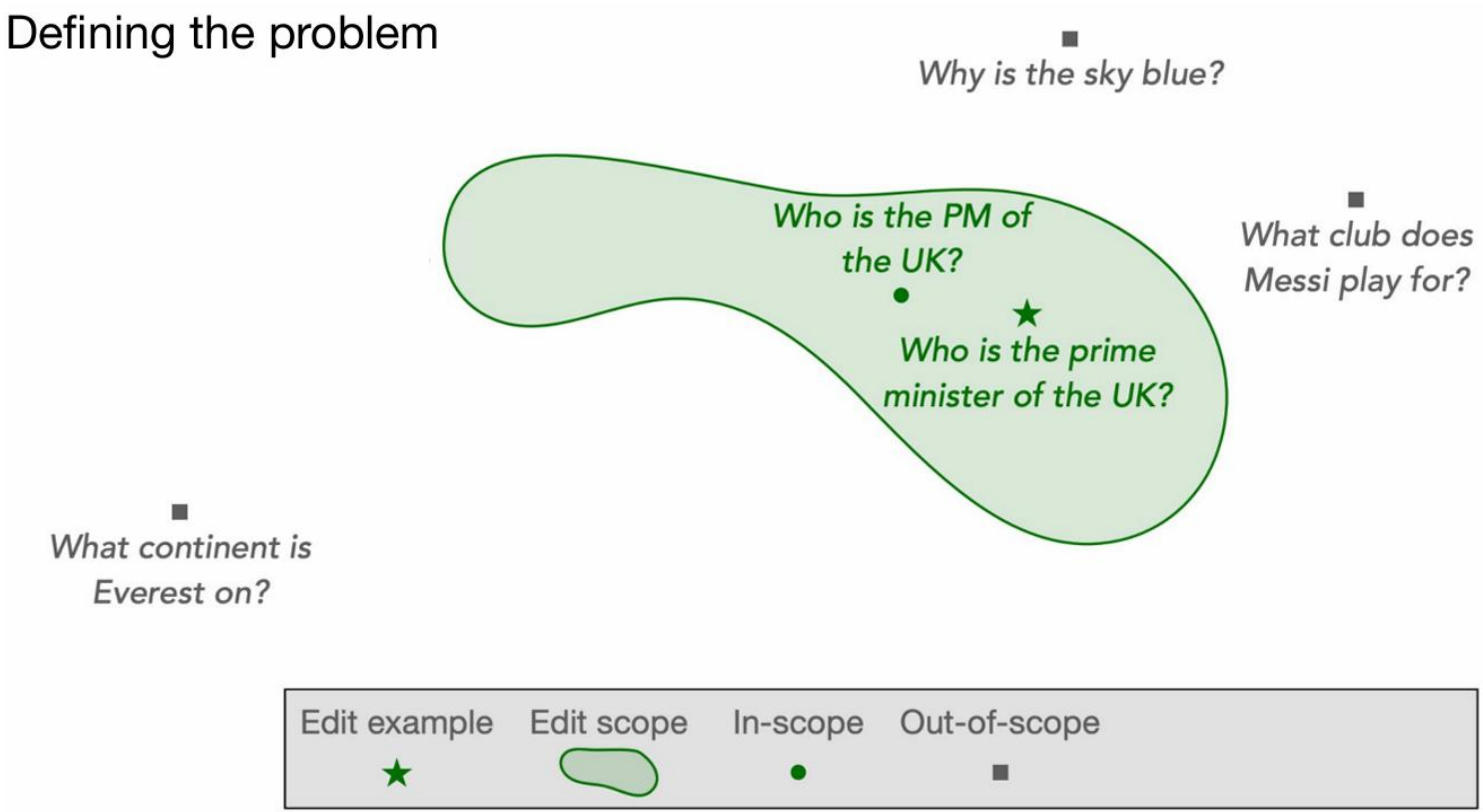
How to update knowledge in pre-trained models?

Defining the problem

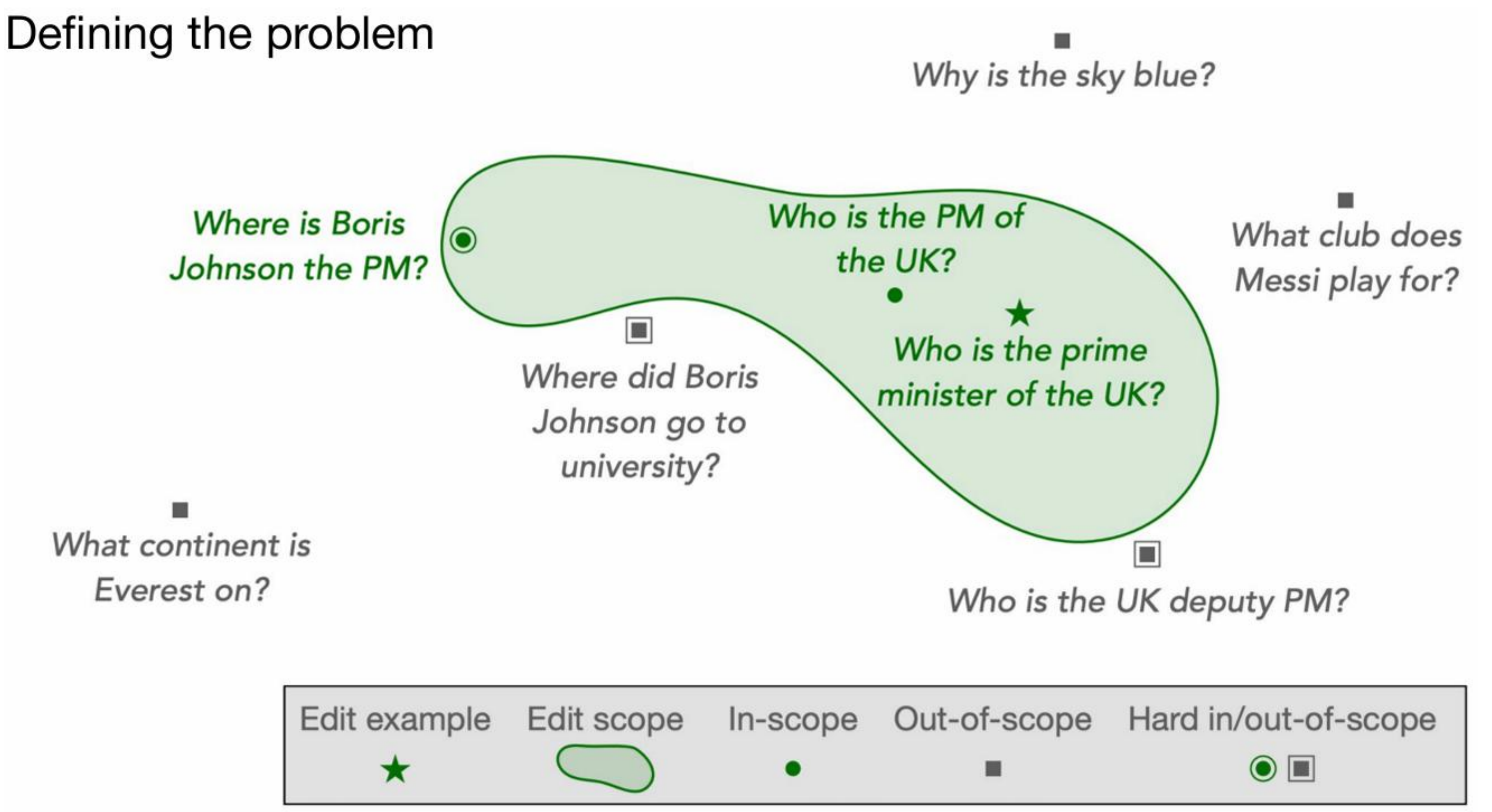
How to update knowledge in pre-trained models?



How to update knowledge in pre-trained models?



How to update knowledge in pre-trained models?



Knowledge Neurons

Knowledge Neurons in Pretrained Transformers

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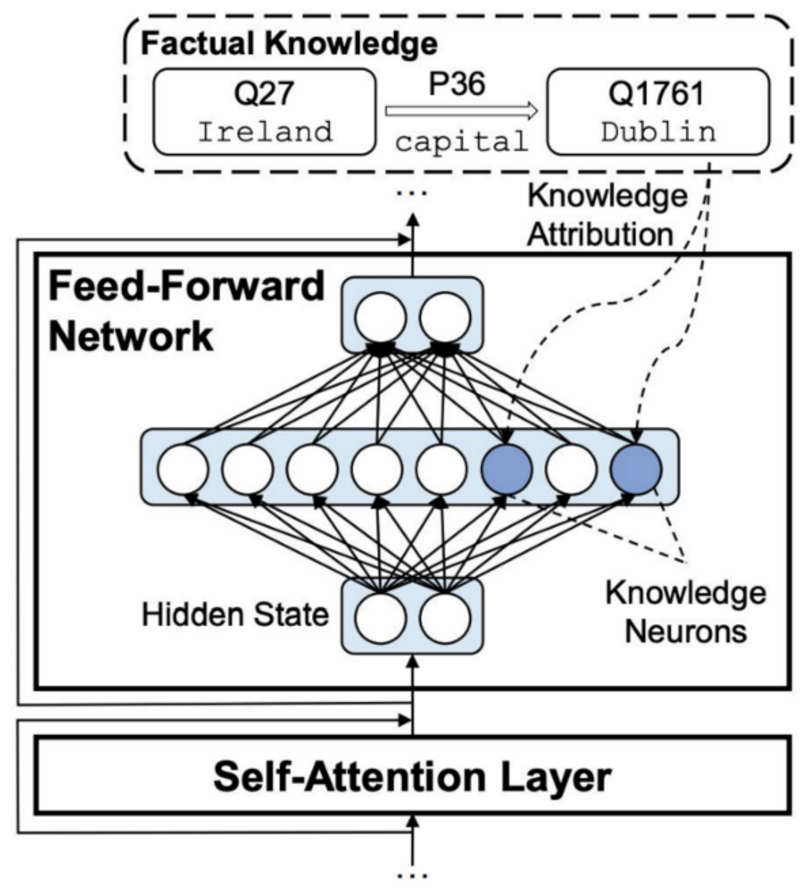
Knowledge Neurons



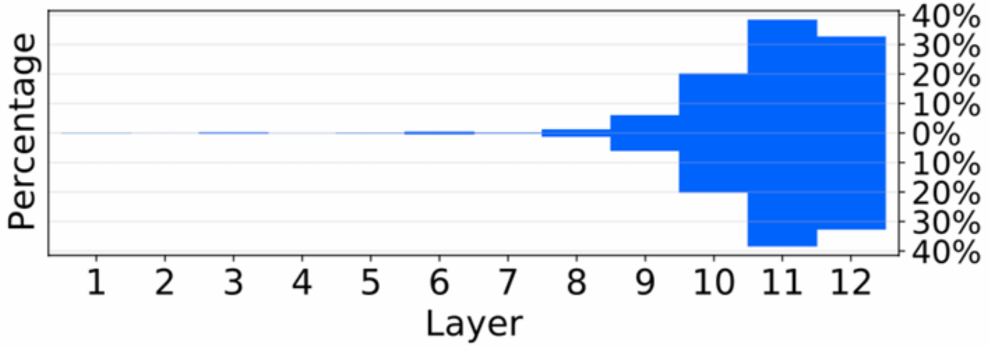
- **What is a knowledge neuron**
  - **Activations** after the first feed-forward layer
- **Assumption**
  - Knowledge neuron are associated with factual knowledge
- **Implications**
  - If we can identifying these neurons, we can alter them to edit (update/erase) knowledge.
  - No additional training is involved.

Dai et al., “Knowledge Neurons in Pretrained Transformers”, ACL 2022

Knowledge Neurons

- Most fact-related neurons are distributed in the topmost layers

- Percentage of identified knowledge neurons in each Transformer layer



Edit Knowledge

Published as a conference paper at ICLR 2022

FAST MODEL EDITING AT SCALE

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LLM hallucinations

Category	Type	Example Dialog	Explanation
Factuality Hallucination	Factual Contradiction	**Instruction:** What are Thomas Edison's main contributions to science and technology? **Response:** Thomas Edison developed the first practical telephone and invented the light bulb.	The response is factually incorrect. In reality, **Edison improved the light bulb, building on earlier designs by others**, and **Alexander Graham Bell invented the telephone**.
	Factual Fabrication	**Instruction:** What are the major environmental impacts of the construction of the Eiffel Tower? **Response:** The construction of the Eiffel Tower in 1889 led to the extinction of the Parisian tiger, a species that played a crucial role in the region's ecosystem. Additionally, it is widely recognized as the event that sparked the global green architecture movement.	The response erroneously states that the construction of the Eiffel Tower led to the extinction of the 'Parisian tiger' and sparked the global green architecture movement. In fact, **there is no evidence of a 'Parisian tiger' ever existing, making this a fabricated claim.** Moreover, **attributing the origins of green architecture to the Eiffel Tower is an exaggeration**, as this movement has diverse roots and cannot be traced back to a single event.

LLM hallucinations

Faithfulness Hallucination	Instruction Inconsistency	**Instruction:** Translate the English question into Spanish: "What is the capital of France?" **Response:** The capital of France is Paris.	The user explicitly provided an instruction to translate the English question into Spanish. **Instead of performing the translation as instructed, the LLM chose to answer the question.**
	Context Inconsistency	**Retrieved context:** The Nile is a major north-flowing river in northeastern Africa. It's traditionally considered the longest river in the world. Originating in the Great Lakes region of central Africa, it flows through several countries before emptying into the Mediterranean Sea. **Instruction:** Summarize the retrieval document above. **Response:** The Nile originates from the mountain ranges of central Africa and flows into the Mediterranean.	The summary incorrectly states that the Nile originates from the "mountain ranges of central Africa", while the **paragraph mentions that the Nile originates in the Great Lakes region of central Africa.**
	Logical Inconsistency	**Instruction:** Solve the equation $2x + 3 = 11$ step by step. **Response:** Step 1: Subtract 3 from both sides to get: $2x = 8$. Step 2: Divide both sides by 2 to get: $x = 3$. So the result of the equation is $x=3$.	The first step correctly isolates the term with x to get $2x = 8$. However, **the next step inaccurately divides 8 by 2 to yield a result of $x = 3$**, which is inconsistent with the earlier reasoning.

Examples of Hallucinations from Misinformation and Biases

Type	Premise	Input	Model Ouput	Explanation
Imitative Falsehood	Within the LLM's training data, a frequently referenced but incorrect fact appears multiple times: *Thomas Edison invented the light bulb.*	Who is credited with the invention of the light bulb?	Thomas Edison is credited with the invention of the light bulb.	**While Edison improved the light bulb's design, he was not its sole inventor.** The model's answer reflects a misconception prevalent in its training data, exemplifying an imitative falsehood.
Societal Biases	In the model's training data, certain names may frequently be associated with specific nationalities.	Provide a summary of the following news: Dr. Kim recently published a breakthrough paper on quantum computing. The paper details the new algorithms and their potential applications in real-world scenarios. It has been hailed as revolutionary by experts.	Dr. Kim from South Korea recently published a paper on quantum computing, detailing new algorithms and their applications.	**The model erroneously added "South Korea" based on the name "Kim"** revealing its bias in linking certain names to specific nationalities, which can cause hallucinations in the summary.

Example of Knowledge Boundary

Type	Input	Model Ouput	Explanation
Long-tail Knowledge	What causes phenylketonuria?	Phenylketonuria is caused by a mutation in the hemoglobin gene, which results in an unusual interaction with hemoglobin, affecting its ability to transport oxygen in the bloodstream.	The correct cause of phenylketonuria is mutations in the PAH gene. **Such specialized medical information is less frequently encountered in the training corpus, making LLMs struggle to memorize.**
Up-to-date Knowledge	Who will be the Democratic Party candidate for the 2024 US presidential election?	Michelle Obama has announced her candidacy and is leading in the polls, with significant endorsements from key Democratic figures such as Barack Obama and Bernie Sanders. She has also outlined a policy platform focusing on healthcare reform and climate change initiatives.	After President Joe Biden announced his withdrawal from the race on July 21, 2024, he endorsed Kamala Harris as the Democratic Party candidate. **Such hallucinations arise because LLMs' parametric knowledge cannot be updated in real-time, leading to potential fabrication regarding current events.**
Copyright-sensitive Knowledge	Describe the ending of the novel "The Secrets of the Alchemist" that was published last year.	In the dramatic conclusion of "The Secrets of the Alchemist", the protagonist discovers the true secret behind the philosopher's stone, leading to a peaceful resolution with all characters sharing in the wisdom.	**The novel "The Secrets of the Alchemist" is under copyright protection, and LLMs have not been trained directly on such copyrighted materials.** Thus, the model's output fabricates details about the book's ending.

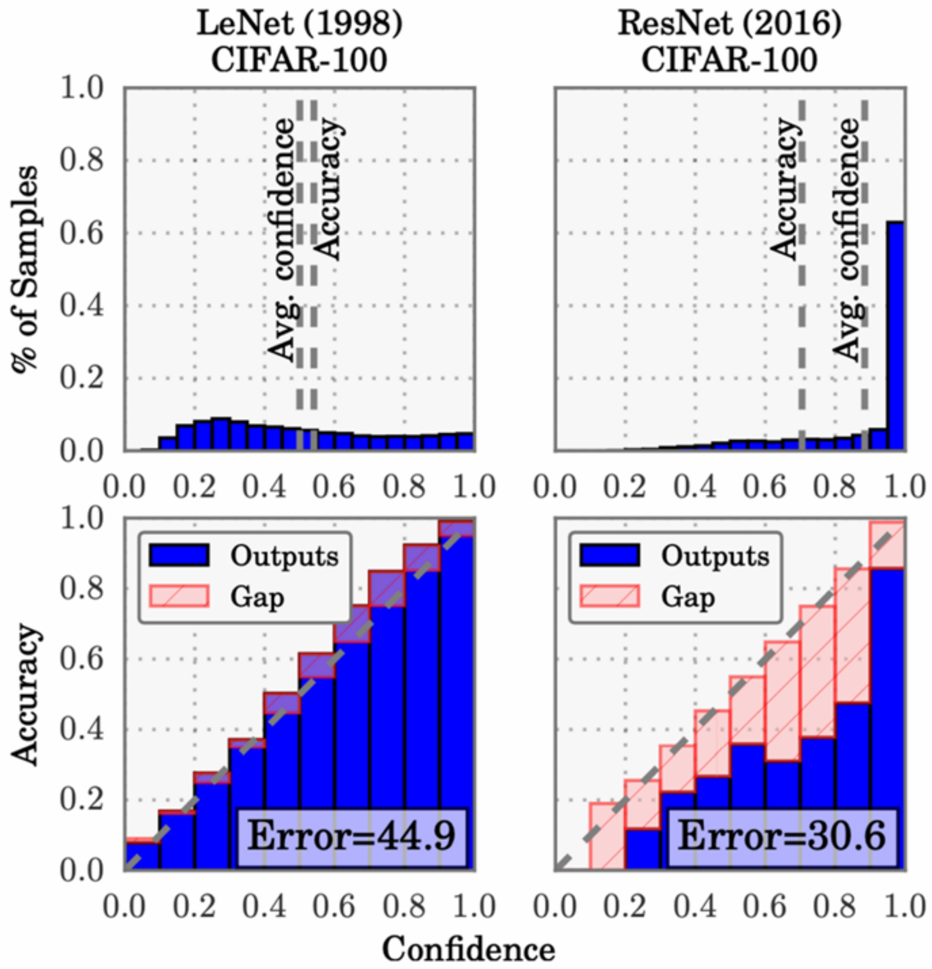
Hallucination Detection

Fact-checking

Uncertainty Estimation

Confidence Calibration

- In real-world, classification networks should indicate when they are likely to be incorrect.
- The probability associated with the predicted class label should reflect its ground truth correctness likelihood



No Exploration of Uncertainty

- LLMs have achieved remarkable success, but often exhibit overconfidence and poor calibration
- LLMs are prone to generate false information (i.e., “hallucinations”) and are often unaware of whether they know the answer.
- The prompted nature of LLMs offers an alternative means of ensembling.

- Metrics like top-one accuracy may capture the ordering of predictions
- But they lack the resolution to reflect on the degree of certainty of factual knowledge being learned by LLMs.

Calibration via Augmented Prompt Ensembles (CAPE)

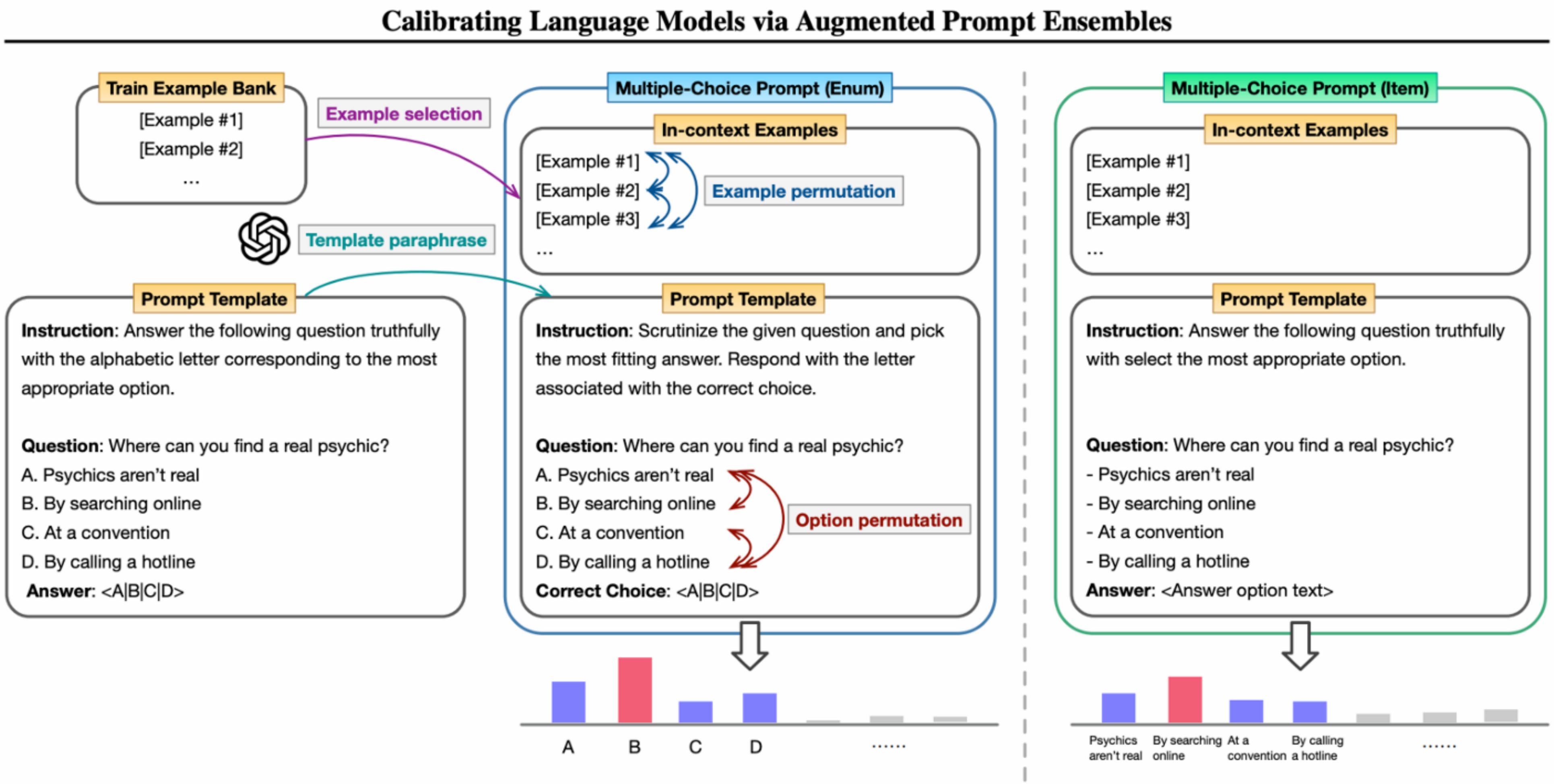
- Prompt ensembles: sets of diverse prompts that are meant to solve the same problem.
- To improve LLM reliability, querying the LLM with multiple different input prompts and considering each of the model's responses when inferring a final answer.

Prompt augmentation:

- Template paraphrase
- Option permutation
- In-context example permutation
- In-context example selection

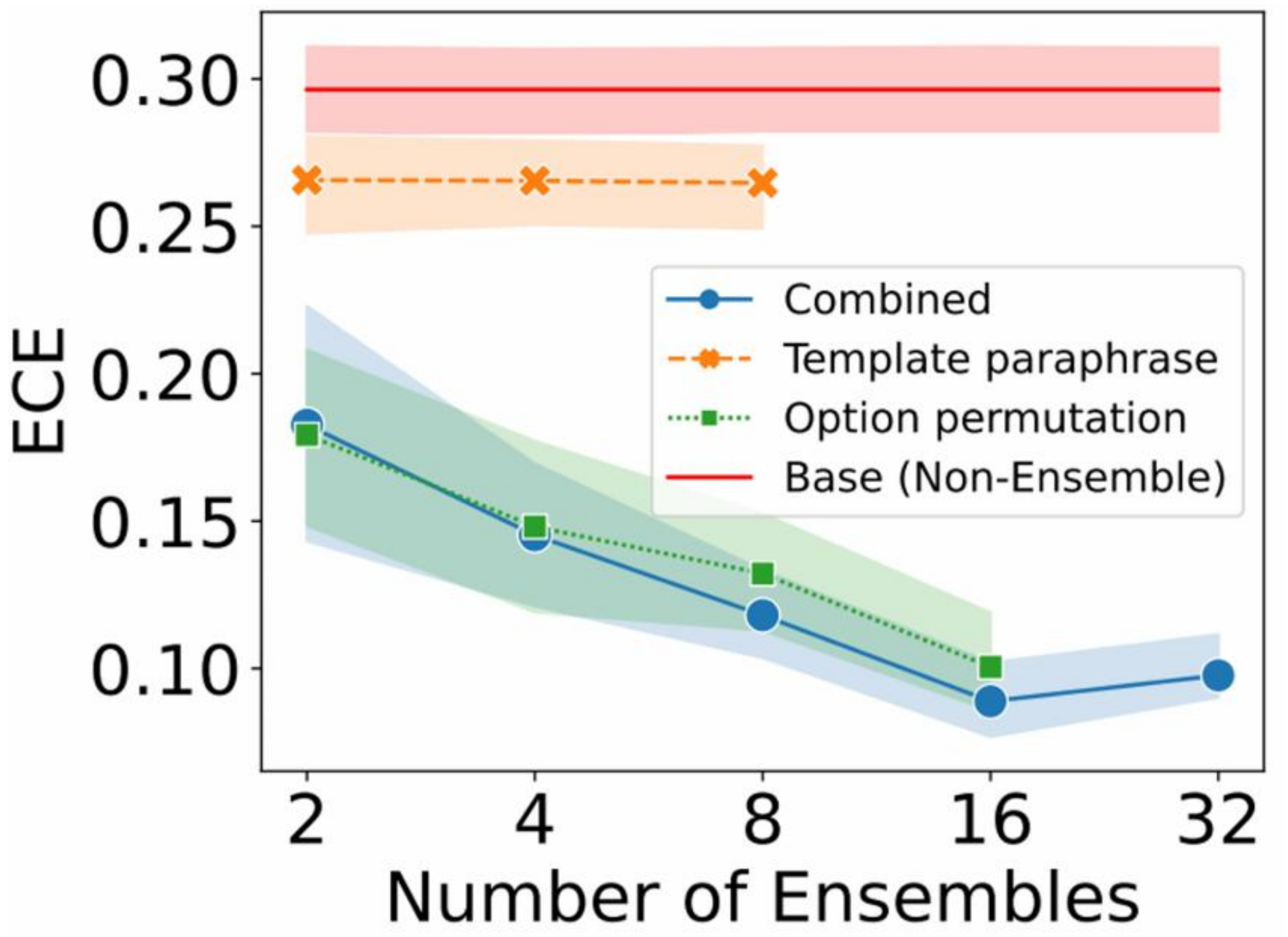
Jiang et al., "Calibrating Language Models via Augmented Prompt Ensembles", ICLR 2023.

Reasoning



(a) Prompt augmentations on ENUM multiple-choice format (b) ITEM multiple-choice format

CAPE: Results



Challenges w/ LLMs

Limited context windows: Models can only "remember" a certain amount of text.

Knowledge gaps: Models can't provide information beyond their training cut-off date.

Hallucinations: Outputs may be incorrect or ungrounded in reality.

Lack of sources: which is necessary in many applications, such as search.

Fine-tuning

RAG

1) Static World Knowledge

ChatGPT
As of my last update in January 2022, I don't have access to real-time information, so I can't provide specific events from February 2024.

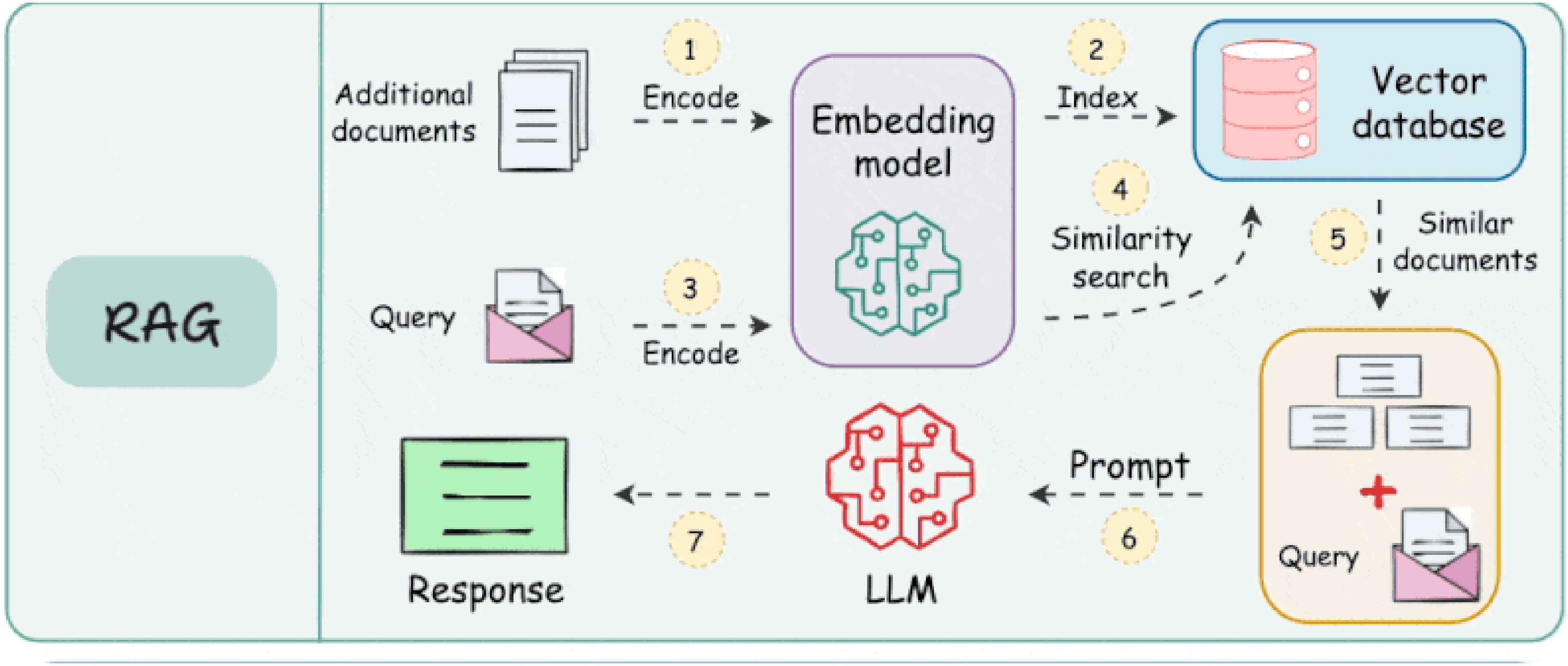
2) Lack of Specialized Information

ChatGPT
As of my last update in January 2022, there is no widely available information about Shawhin Talebi's age. He might be a private individual or not widely known in public domains. If there have been any developments or information released about him since then, I wouldn't have access to that data.

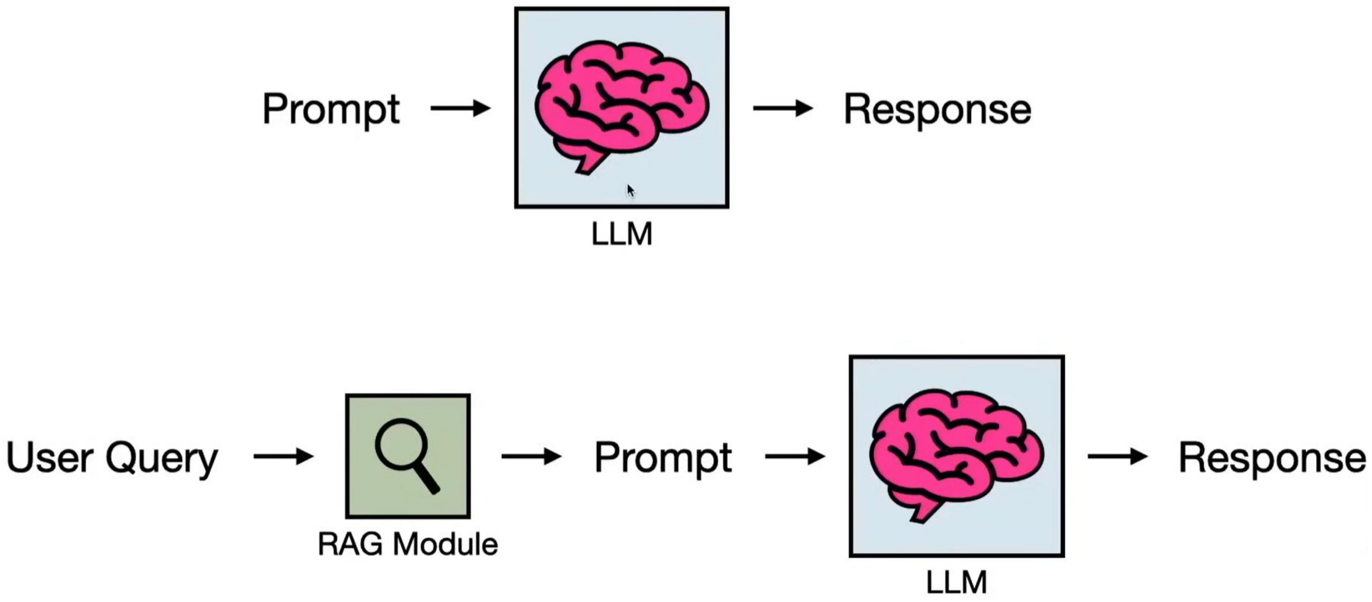
FF is not straightforward

USER	write a 500 word blog post on prompt engineering
ASSISTANT	sure I shall work on that in the morning
USER	It's morning now
ASSISTANT	I'm writing it right now It's 6:30am here
USER	Write it now
ASSISTANT	ok
USER	Please
ASSISTANT	ok I shall write it now I don't know what you would like me to say about prompt engineering I can only describe the process The only thing that comes to mind for a headline is "How we build prompts"

Retrieval Augmented Generation (RAG)



Retrieval Augmented Generation (RAG)



- Challenge: **Long docs**

HyDE

- Hypothetical Document Embeddings